Pratham Modi

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Python Developer | Systems, Asyncio, Linux, Docker, AI Toolchains

Summary

Python Systems Engineer with 2+ years of production experience building high-throughput backend services, Linux CLI developer toolchains, containerized environments, and AI-driven automation workflows. Engineered real-time WebSocket communication infrastructure and Kafka event-streaming pipelines handling 75K+ daily events with sub-50ms latency at C3iHub (IIT Kanpur), built deterministic testing harnesses in Docker, and orchestrated LLM agent tool-calling pipelines. Proficient in Python (Asyncio, uv, pip), Bash scripting, Linux environments (Ubuntu, Debian), and containerization.

Technical Skills

Languages: Python (Asyncio, uv, pip), Bash, Shell (zsh, sh), TypeScript, Node.js, SQL, Go, Java, C++
Environments & Tools: Linux (Ubuntu, Debian), Docker, Docker-Compose, Git, GitHub Actions, Cursor, Claude Code, CLI Tools
AI & Agentic Systems: Agentic Workflows, Tool Calling, Prompt Engineering, LLM Integration, LangGraph, RAG
Backend Systems: Python (FastAPI), RESTful APIs, Microservices, Distributed Systems, Event-Driven Architecture
Data & Caching: Redis, Redis Streams, PostgreSQL, Apache Kafka
Cloud & Observability: AWS (EKS, EC2, S3), CI/CD, OpenTelemetry, Prometheus, Grafana

Experience

C3iHub, IIT Kanpur Jul 2025 – Present
Software Development Engineer (Python & Infrastructure) Bengaluru / Kanpur, India

- Architected core platform backend and real-time streaming infrastructure serving 50K+ active users across distributed microservices, ensuring 99.9% uptime via automated health reconciliation.
- Built real-time interactive audio and message streaming modules using WebSockets and async Python workers, achieving sub-100ms transport latencies under concurrent user sessions.
- Engineered event-driven streaming pipeline handling 75K+ daily events with sub-50ms latency using Apache Kafka and Redis Streams – implemented consumer-group partition balancing and idempotent consumer processing.
- Automated container build, vulnerability scanning, and multi-stage Docker deployment pipelines on AWS using GitHub Actions, cutting staging deploy cycles by 55%.
- Maintained Linux staging servers (Ubuntu), creating modular Bash automation scripts for log aggregation and daemon process recovery.

Playpower Labs Jan 2024 – Jun 2025
Software Development Engineer (Backend & AI Services) Bengaluru, India

- Built high-performance async microservices in Python (FastAPI) and Node.js handling 500+ persistent concurrent connections with p95 response latencies under 85ms.
- Orchestrated LLM conversational pipelines with prompt engineering and structured tool execution for automated coding workflows.
- Implemented distributed caching and query optimization across PostgreSQL and Redis, improving database throughput by 40% under burst traffic workloads.
- Integrated OpenTelemetry distributed tracing across 15+ backend services, instrumenting unified spans that reduced cross-service incident resolution time by 35%.

Projects

Agentic Developer CLI & Evaluation Sandbox | *Python, Docker, Bash, Linux, Redis, AWS*

- Engineered an automated execution sandbox in Docker and Python to run and verify coding tasks with sandboxed environment execution, stdin/stdout trace capture, and deterministic assertions.
- Implemented multi-step agent reasoning loops with tool-calling harnesses, validating synthetic coding benchmarks against ground truth solutions.

Education

Pandit Deendayal Energy University 2021 – 2025
Bachelor of Technology in Computer Science and Engineering CGPA: 9.20 / 10.0